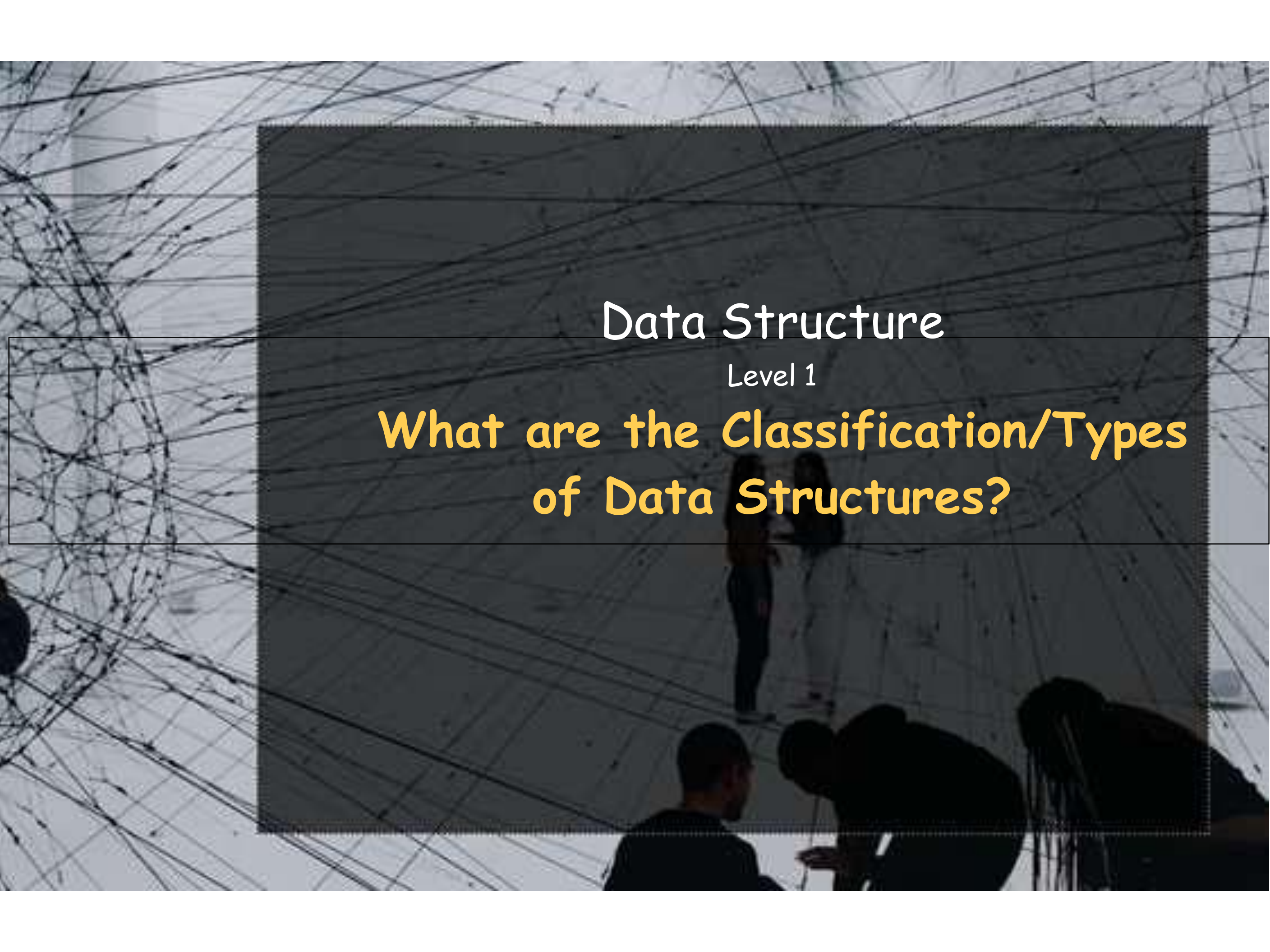


Data Structure

Level 1

What are the
Classification/Types of
Data Structures?

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

Data Structure

Level 1

**What are the Classification/Types
of Data Structures?**

Classification of Data Structurefi

Primitive (Basic) Data Structure.

- Integer
- Float
- Char
- Pointer

Non-Primitive (Advance) Data Structure.

- Linear List
- Non-Linear List

Differences:

A primitive data structure:

- Is generally a basic structure that is usually built into the language, such as an integer, a float.

A non-primitive data structure:

- Is built out of primitive data structures linked together in meaningful ways, such as a or a arrays, linked-list, binary search tree, Tree, graph etc.

Primitive Data Structure:

Primitive
Data Structure.

- Integer
- Float
- Char
- Pointer

- They are basic structures and directly operated upon the machine instructions.
- Integer, Float, Char, pointers..etc., fall in this category.

non-Primitive Data Structure:

Linear List

- Array
- Linfied List
- Stacfi
- Queue

Non-Linear List

- Tree
- Graph

- Complex/Sophisticated Data Structure derived from primitive data structure.
- Emphasize on structuring of group of homogeneous (same type) or heterogeneous (different type) data items.
- The design of an efficient data structure must tafie operations to be performed on data structure.

Linear vs Non-Linear Data Structures

Linear Data Structure

- Data structure in which data elements are arranged sequentially or linearly, where each element is attached to its previous and next adjacent elements, is called a linear data structure.
- Examples of linear data structures are array, stack, queue, linked list, etc.

Non-Linear Data Structure

- Data structures where data elements are not placed sequentially or linearly are called non-linear data structures. In a non-linear data structure, we can't traverse all the elements in a single run only.
- Examples of this data structure are Tree, Graph, etc.

Static vs Non-Static Data Structures

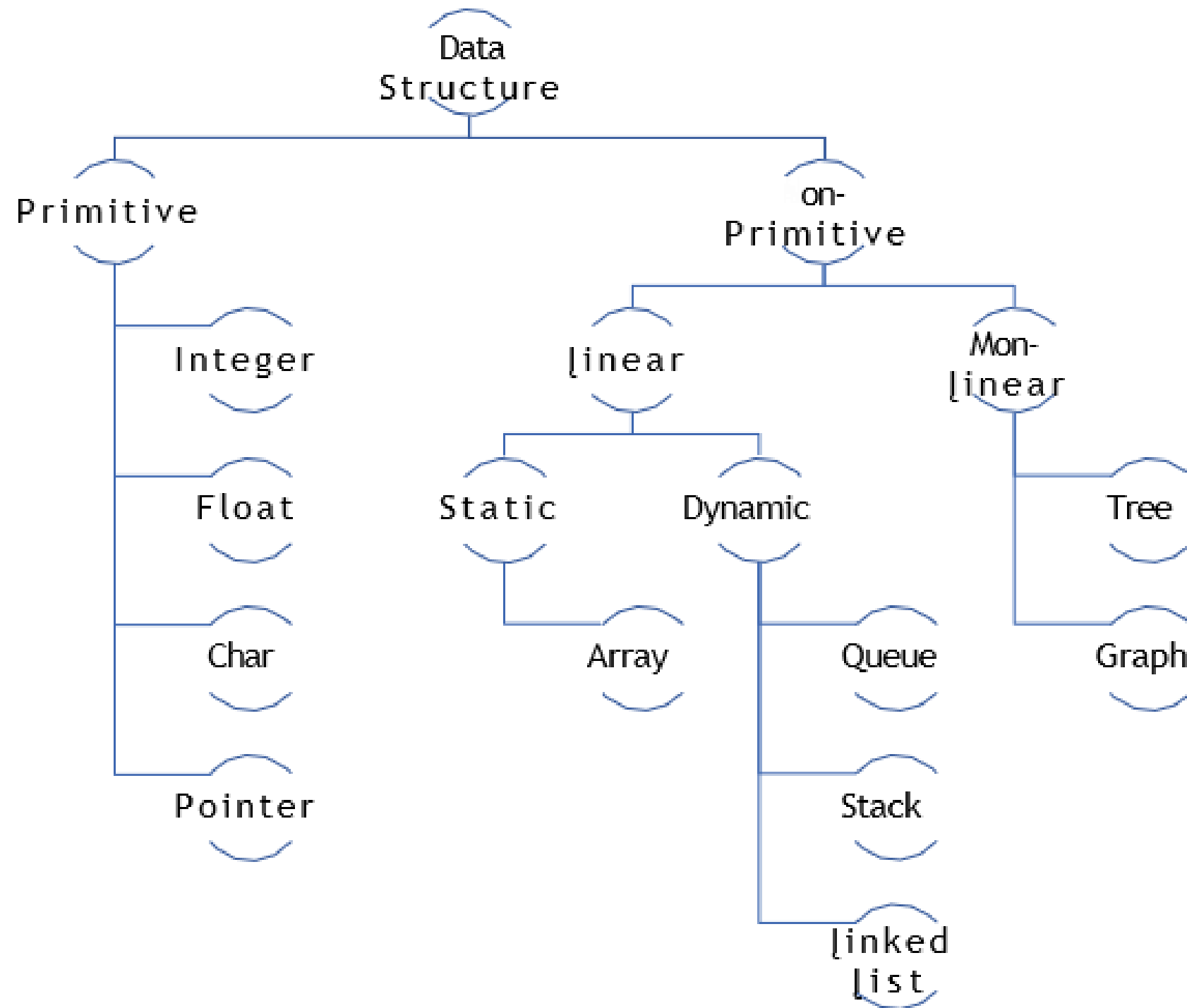
Static data structure

- Static data structure has a fixed memory size. It is easier to access the elements in a static data structure.
- An example of this data structure is an array.

Dynamic data structure

- In dynamic data structure, the size is not fixed. It can be randomly updated during the runtime which may be considered efficient concerning the memory (space) complexity of the code.
- Examples of this data structure are Stack , Queue, etc.

Classifications of Data Structure:



Common Operations on Data Structure:

- Operations On Data

- Create
- Update
- Search
- Select
- Sorting
- Merging
- Destroy or Delete

Thank you 😊